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1 Compactness

We will now discuss the equivalence of topological and sequential compactness in the reals and in general metric spaces. Let us view the following theorem:

1.1 Limit Points

The limits of sequences ARE NOT necessarily limit points. Given some subset $E \subseteq \mathbb{R}$, x is a limit point of E if:

$$\forall \epsilon > 0 : B(x, \epsilon) \cap E \neq \emptyset$$

Notice how this only requires the intersection to be nonempty; which means there could possibly be a finite number of points in the intersection (but we will see this is not possible). Let us look at the following lemma:

Lemma 1.1. *x is a limit point of E if and only if*

$$\exists \{x_n\}_{n \in \mathbb{N}} \in (E \setminus \{x\})^{\mathbb{N}} : x_n \rightarrow x$$

1.2 Sequential Compactness

Theorem 1.1 (Sequential Bolzano Weierstrass). *Any bounded sequence has a convergent subsequence.*

To prove this fact, we show first that any sequence in $\mathbb{R}$ has a monotonic subsequence.

Proof. Let us define the following set, we call "peaks" of a sequence:

$$P := \{n \in \mathbb{N} : (\forall j > n : x_j < x_n)\}$$

In the case that P is countably infinite, then we set:

$$n_0 := \inf P \wedge \forall k > 0 : n_k := \inf (P \setminus \{n_0, \dots, n_{k-1}\})$$

In which case we are guaranteed that $\forall k \in \mathbb{N} : n_{k+1} > n_k$ which tells us that $x_{n_{k+1}} < x_{n_k}$, showing that there exists a strictly decreasing subsequence. In the case that P is finite, we choose the following:

$$n_0 := \sup(P) + 1$$

that way the peaks are fully omitted from the subsequence, as they will prove to be problematic in the construction. Then for any $k \in \mathbb{N}$, we are guaranteed that $n_k \notin P$ (since $\{n_k\}_{k \in \mathbb{N}}$ is strictly increasing and $n_0 \notin P$), which means:

$$\exists j > n_k : x_j \geq x_{n_k}$$

So we simply define:

$$\forall k \in \mathbb{N} : n_{k+1} := \inf \{j > n_k : x_j \geq x_{n_k}\}$$

in which case we obtain a non-decreasing subsequence. Now any bounded sequence has a monotone subsequence, which converges. What this means is given some sequence $\{x_n\}_{n \in \mathbb{N}}$, we have that:

$$\exists y \in \mathbb{R}, \exists r > 0, \forall n \in \mathbb{N} : x_n \in B(y, r)$$

so the sequence has a convergent subsequence within the ball, but such a subsequence may not converge within the ball. This is due to the topological idea that the limit point certainly belongs in the closure of the set but not necessarily in the open set. Consider the bounded sequence:

$$\{x_n\}_{n \in \mathbb{N}} \in (0, 1)^{\mathbb{N}} : \left(\forall n \in \mathbb{N} : x_n := \frac{1}{n+2} \right)$$

Then $x_n \rightarrow 0$ yet $0 \notin (0, 1)$. This idea of closedness plays a key role in sequential compactness. $\square$

Definition 1.1. A sequentially compact interval of $\mathbb{R}$ is one for which every sequence has a convergent subsequence that converges to a point within the interval.

The previous Theorem 1.1 leads to a key theorem:

Theorem 1.2. An interval of $\mathbb{R}$ is sequentially compact if and only if it is closed and bounded.

Proof. $\Leftarrow$: This direction follows from Sequential Bolzano Weierstrass. If the interval is bounded, then any sequence has a convergent monotone subsequence. Since it is closed, the limit of the sequence, which is an adherent point, belongs in the closure.

$\Rightarrow$: Consider any convergent sequence $\{x_n\}_{n \in \mathbb{N}}$ such that $\forall n \in \mathbb{N} : x_n \in I$. Then by sequential compactness:

$$\exists \{n_k\}_{k \in \mathbb{N}} : x_{n_k} \rightarrow x \wedge x \in I$$

which forces $x_n \rightarrow x$. Thus all convergent sequences converge to points within the set (which by sequential closure) means I is closed. Now assume the interval is not bounded, in which case:

$$\forall r > 0, \forall y \in \mathbb{R} : I \not\subseteq B(y, r)$$

Then consider some sequence $\{x_n\}_{n \in \mathbb{N}} \in I^{\mathbb{N}}$. For this, we need the following notion: □

Lemma 1.2. For any set $I \subseteq \mathbb{R}$, I is bounded if and only if it is totally bounded (ONLY for $\mathbb{R}$).